

RALA Formula Lab

Experiment with safe formulas for κ and ϕ in Rank-Augmented Linear Attention.

$\kappa(x)$ is valid.

$\phi(x)$ is valid. Uses `Linear(x)`.

Results are observations for this dataset, seed, and training budget. Higher rank can indicate richer features, but it does not prove better accuracy by itself.
Rank tolerance: $1e-05$ · Stats averaged over 3 batch(es) · 3 warmup pass(es).

Run experiment

Results: RALA

Val accuracy

0.399

Val loss

1.656

Output rank ratio

1.000

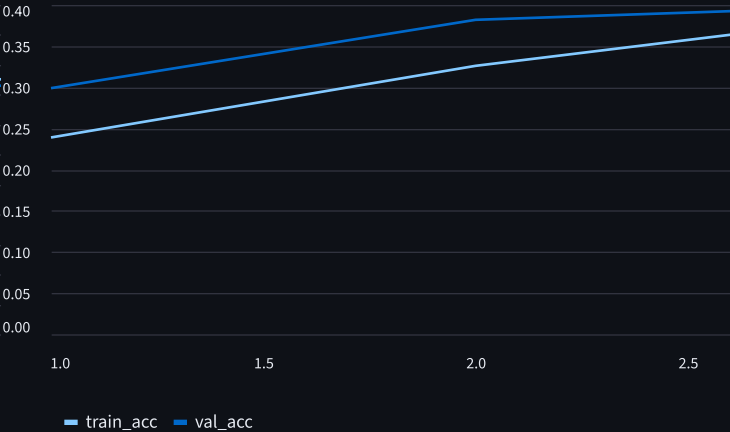
Inference time

295.2 ms

Loss



Accuracy



Per-Layer Rank Diagnostics

Rank tolerance: $1.0e-05$ · Stats averaged over 3 validation batch(es)

	Layer	KV Rank	KV Rank Ratio	Output Rank	Output Rank Ratio	α Sum Mean
0	0	15.998	1.000	16.000	1.000	256.000
1	1	15.997	1.000	16.000	1.000	256.000

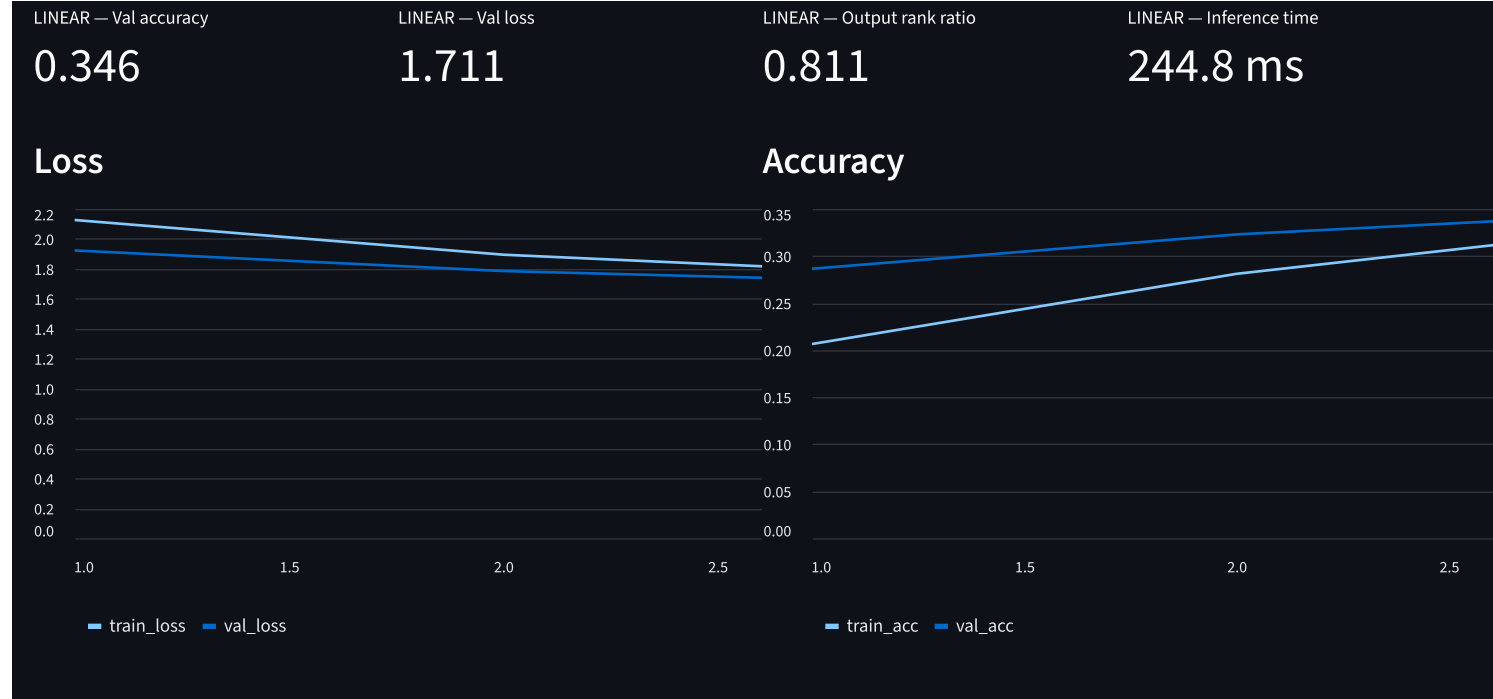
Bottleneck Summary (worst-case layer)

	Metric	Value
0	Bottleneck layer	
1	KV buffer rank	
2	KV rank ratio	
3	Output rank	
4	Output rank ratio	
5	Alpha sum mean	
6	Min denominator	
7	Inference time (ms)	

No NaN, Inf, large-activation, or near-zero-denominator warnings found.

Baseline Comparison

Baseline: LINEAR



Per-Layer Rank Diagnostics

Rank tolerance: 1.0e-05 · Stats averaged over 3 validation batch(es)

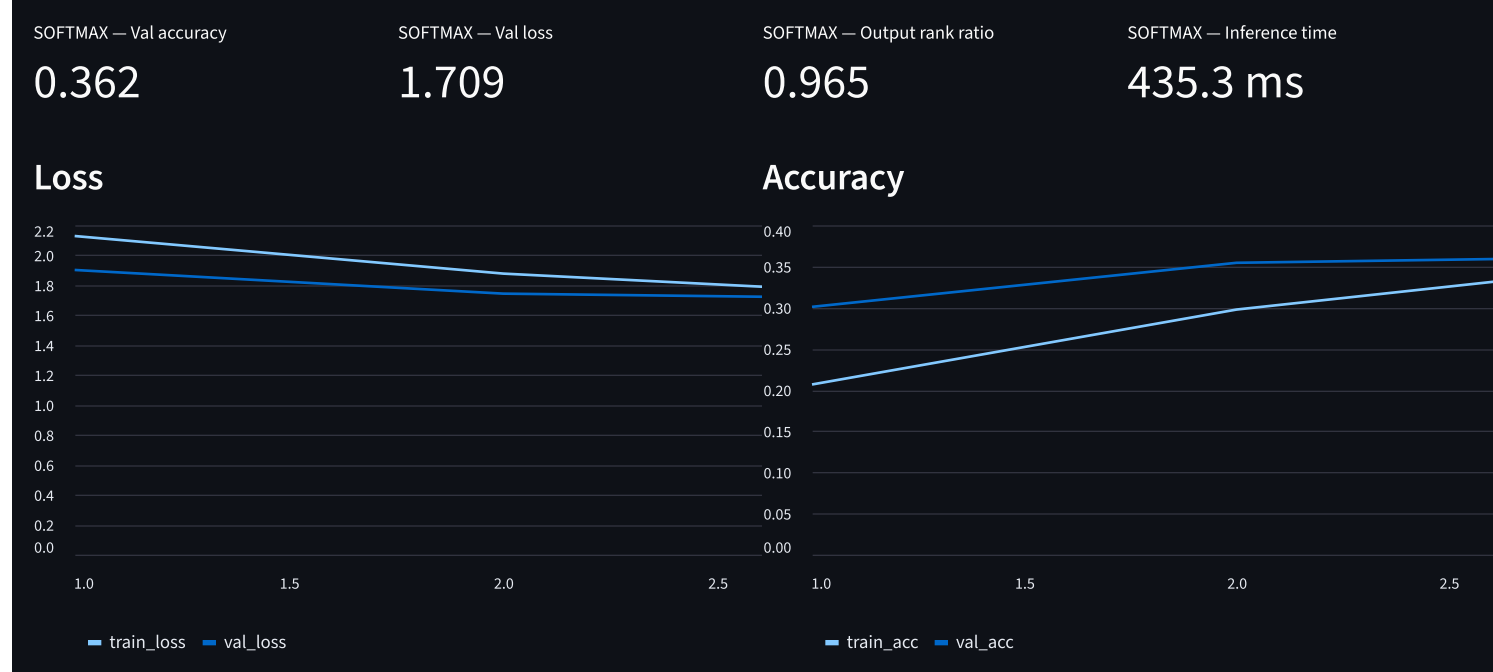
	Layer		KV Rank	KV Rank Ratio	Output Rank	Output Rank Ratio	α Sum Mean
	0	0	15.999	1.000	12.973	0.811	256.000
	1	1	15.997	1.000	13.277	0.830	256.000

Bottleneck Summary (worst-case layer)

	Metric	Value
0	Bottleneck layer	
1	KV buffer rank	
2	KV rank ratio	
3	Output rank	
4	Output rank ratio	
5	Alpha sum mean	
6	Min denominator	
7	Inference time (ms)	

No NaN, Inf, large-activation, or near-zero-denominator warnings found.

Baseline: SOFTMAX



Per-Layer Rank Diagnostics

Rank tolerance: 1.0e-05 · Stats averaged over 3 validation batch(es)

	Layer	KV Rank	KV Rank Ratio	Output Rank	Output Rank Ratio	α Sum Mean
0	0	n/a	n/a	15.979	0.999	n/a
1	1	n/a	n/a	15.439	0.965	n/a

Bottleneck Summary (worst-case layer)

	Metric	Value
0	Bottleneck layer	
1	KV buffer rank	
2	KV rank ratio	
3	Output rank	
4	Output rank ratio	
5	Alpha sum mean	
6	Min denominator	
7	Inference time (ms)	

No NaN, Inf, large-activation, or near-zero-denominator warnings found.

Side-by-Side Summary

	Attention	Val Accuracy	Val Loss	Output Rank Ratio	KV Rank Ratio	Inference (ms)
0	RALA	0.399	1.656	1.000	1.000	295.2
1	LINEAR	0.346	1.711	0.811	1.000	244.8
2	SOFTMAX	0.362	1.709	0.965	n/a	435.3

Metric guide

Accuracy. Fraction of correct predictions. Higher is better, but compare repeated seeds before making claims.

Loss. Cross-entropy training or validation objective. Lower is better.

KV buffer rank. Average rank of the global $B = \sum \alpha_k(K)^T V$ memory. Higher suggests richer global memory.

Output rank. Average rank of the attention output feature map. Higher suggests more diverse token features.

Rank ratio. Rank divided by the maximum possible rank. Values closer to 1.0 are closer to full-rank.

Inference time. Forward-pass wall-clock time for one validation batch. Lower is faster.

Stability. Warnings for NaN, Inf, large activations, or near-zero denominators.

Download experiment JSON